7800 - An Empirical Benchmark for H2 and PAHs at Extremely Low Metallicity

Cycle: 4, Proposal Category: GO

INVESTIGATORS

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Dr. Grace Telford (CoI)	University of Utah
Dr. Ryan Rickards Vaught (CoI)	Space Telescope Science Institute

OBSERVATIONS

Folder	*Observation*	*Label*	*Observing Template*	*Science Target*
Observation Folder				
	1	LeoP-NIRspec	NIRSpec IFU Spectroscopy	(1) LEO-P
	2	LeoP-OFF-NIRSpec	NIRSpec IFU Spectroscopy	(2) LEO-P-OFF

ABSTRACT

JWST has revealed that high-redshift galaxies can form stars at a much higher rate than previously thought. Since the far distances of these early galaxies prohibits detailed study, observations of nearby, metal-poor galaxies that mimic the pristine gas conditions of these distance galaxies are needed to constrain models of star formation and dust content. Until recently, both the observations of small dust grains known as polycyclic aromatic hydrocarbons (PAHs) and the primary tracer of molecular gas, CO, have been difficult or impossible to detect in low metallicity galaxies. Now with the resolution and sensitivity of JWST, we can detect and characterize this faint ISM emission. Recent MIRI-MRS observations of the closest, extremely metal-poor (3% Solar) galaxy Leo P have directly detected molecular hydrogen through the MIR rotationally excited transition S(1). We propose deep NIRSpec observations centered on the H2 emission to 1) detect the vibrational, FUV-pumped H2 lines in the NIR to constrain the temperature and mass of the warm molecular gas and 2) target the 3.3 micron PAH complex to detect or put a stringent upper limit on the presence of PAHs at 3% Solar metallicity. The requested observations provide the best chance to ever detect PAHs in Leo P due to the strong correlation between the warm H2 and PAHs. We will analyze the H2 and PAHs together, which will determine the level of destruction of the small grains, identify the main heating sources of the H2 gas, and constrain the H2 mass. The characterization of the ISM in Leo P will be tied to star formation models of high redshift and local galaxies, revolutionizing our picture of the metal poor ISM.

OBSERVING DESCRIPTION

We request deep, NIRSpec observations centered on the MIRI-MRS detection of H2 S(1) rotationally excited emission in the metal-poor galaxy Leo P. The G395M/290LP is chosen to capture the 3.3 micron PAH complex and the G235M/F170LP gratings will cover the 2.12 micron vibrational line, as well as the vibrational cascade in the near-infrared (including the 2.22, 2.03, and the 1.96 micron H2 lines). We use the NRSIRS2 readout pattern, which is optimized for long exposure IFU observations of faint targets and minimizes data volume. For the G395M/290LP grating targeting the PAHs, we request 20 groups per integration, 4 integrations, and use the 4-POINT-DITHER for extended sources for a total of 6.5 hours. For the estimated PAH and H2 signal, this integration time will be achieve an SNR = 12 for the 3.3 micron feature, SNR = 5 for the 3.4 micron feature. We require an uninteruptable dedicated background for the G395M/290LP targeting PAH emission, as the features will be faint and broad. For the G235M/F170LP grating, we request 15 groups per integration, 2 integrations, and use the 4-POINT-DITHER for extended sources for a total of 2.5 hours, achieving an SNR = 15 for the 2.12 micron H2 vibrational transition. A dedicated background is not required for the vibrational H2 lines in the G235M/F170LP gratings, as these features will be narrow and brighter. We do not need a dedicated MSA leakcal exposure because the MSA will fall on a faint part of the sky outside far from the target. These observations do not require target acquisition, as the location of the target is well characterized and the MIRI-MRS observations were successful without a dedicated TAQ.

Fixed Targets

#	Name	Target Coordinates	Targ. Coord. Corrections	Miscellaneous
(1)	LEO-P	RA: 10 21 45.1076 (155.4379483d) Dec: +18 05 17.46 (18.08818d) Equinox: J2000	Epoch of Position: 2024.5	
Comments: This object was generated by the targetselector and retrieved from the SIMBAD database. Category=Galaxy Description=[Dwarf galaxies, Dwarf irregular galaxies] Extended=YES				
(2)	LEO-P-OFF	RA: 10 21 41.8460 (155.4243583d) Dec: +18 05 1.02 (18.08362d) Equinox: J2000	Epoch of Position: 2000	
Comments: This object was generated by the targetselector and retrieved from the SIMBAD database. Category=Unidentified Description=[Blank field] Extended=NO				

Observation

Proposal 7800, Observation 1: LeoP-NIRspec

Mon Feb 23 14:00:11 GMT 2026

Diagnostic Status: Warning

Observing Template: NIRSpec IFU Spectroscopy

Diagnostics

(Visit 1:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.

Fixed Targets

#	Name	Target Coordinates	Targ. Coord. Corrections	Miscellaneous
(1)	LEO-P	RA: 10 21 45.1076 (155.4379483d) Dec: +18 05 17.46 (18.08818d) Equinox: J2000	Epoch of Position: 2024.5	

Comments: This object was generated by the targetselector and retrieved from the SIMBAD database.
Category=Galaxy
Description=[Dwarf galaxies, Dwarf irregular galaxies]
Extended=YES

Template

TA Method	HFF Readout Mode
NONE	false

Dithers

#	Dither Type	Size	Starting Point	Number of Points	Points
1	4-POINT-DITHER				

Spectral Elements

#	Grating/Filter	Readout Pattern	Groups/Int	Integrations/Exp	Leakcal	Dither	Autocal	Total Dithers	Total Integrations	Total Exposure Time	Optional ETC ID
1	G395M/F290LP	NRSIRS2	20	4	false	true	NONE	4	16	23575.646	222975
2	G235M/F170LP	NRSIRS2RAPID	75	2	false	true	NONE	4	8	8870.045	222975

Special Requirements

Group Observations 1, 2, Non-interruptible

Observation

Proposal 7800, Observation 2: LeoP-OFF-NIRSpec

Mon Feb 23 14:00:11 GMT 2026

Diagnostic Status: Warning

Observing Template: NIRSpec IFU Spectroscopy

Diagnostics

(Visit 2:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.

Fixed Targets

#	Name	Target Coordinates	Targ. Coord. Corrections	Miscellaneous
(2)	LEO-P-OFF	RA: 10 21 41.8460 (155.4243583d) Dec: +18 05 1.02 (18.08362d) Equinox: J2000	Epoch of Position: 2000	

Comments: This object was generated by the targetselector and retrieved from the SIMBAD database.
Category=Unidentified
Description=[Blank field]
Extended=NO

Template

TA Method	HFF Readout Mode
NONE	false

Dithers

#	Dither Type	Size	Starting Point	Number of Points	Points
1	4-POINT-DITHER				

Spectral Elements

#	Grating/Filter	Readout Pattern	Groups/Int	Integrations/Exp	Leakcal	Dither	Autocal	Total Dithers	Total Integrations	Total Exposure Time	Optional ETC ID
1	G395M/F290LP	NRSIRS2	20	4	false	true	NONE	4	16	23575.646	222975

Special Requirements

Group Observations 1, 2, Non-interruptible